A hybrid quantum-classical campaign

Hybrid Quantum-Classical Term Deposit Campaign Pipeline

- Bank term-deposit campaign: predict who subscribes, pick 6 customers

to contact under budget + capacity limits, estimate the chance at

least 3 of them subscribe.

- FINAL ANSWER — contact: C0031, C0079, C0256, C0338, C0170, C0327
- P(at least 3 of 6 subscribe) — quantum 0.3574 | exact 0.3603 | Monte Carlo 0.3595

- Solo entry — Umar Shareef, UAF BSCS. Qiskit + Aer simulator, seed 42.
- All three stages are linked: Stages 2 and 3 use the Stage 1 probabilities.

The pipeline: three linked stages

One connected system — errors carry forward by design

- STAGE 1 — PREDICT: 300 labelled customers train a hybrid classifier; probabilities for all 100 test customers. Top 10 move on.
- STAGE 2 — SELECT: QAOA picks exactly 6 of the top 10, maximising expected net value under budget (25) and capacity (22).
- STAGE 3 — SIMULATE: one qubit per selected customer, RY angle set from its probability; count outcomes with at least 3 ones.

- Quantum vs classical is separated at every stage, and every result is checked against an exact classical reference.

Stage 1 — hybrid classifier (5-fold CV, train only)

Model	ROC-AUC	PR-AUC	Brier	Log loss
Logistic regression	0.6286	0.2001	0.2025	0.5955
Random forest	0.6467	0.2331	0.1011	0.3555
Gradient boosting	0.5663	0.1924	0.114	0.524
Quantum kernel SVC	0.4905	0.1204	0.1034	0.3622
Hybrid (logreg + Q-kernel)	0.6293	0.1972	0.1228	0.4093
Hybrid (RF + Q-kernel)	0.6437	0.2221	0.1003	0.3474

Final submitted model: Random forest — chosen by the pre-registered OOF PR-AUC rule (see README).
Base subscription rate in train: 11.7% (35/300).
Quantum kernel: ZZFeatureMap (4 qubits, reps 2) on pdays, campaign, balance, day —
features chosen by train-only mutual information. Preprocessing fit on train only.

Stage 2 — QAOA selection

Exactly 6 of the top 10, under budget and capacity

- Candidates: top 10 by Stage 1 probability — C0031, C0079, C0081, C0256, C0338, C0183, C0170, C0378, C0327, C0188
- Selected (QAOA): C0031, C0079, C0256, C0338, C0170, C0327
- Objective (expected net value): QAOA 33.22 | brute-force optimum 33.22 | gap 0.0000
- Totals: contact cost 10 / 25 • capacity 15 / 22 • exactly 6 selected

- QUBO with penalty terms for the three constraints; decoded from measured bitstrings, feasibility checked on every sampled string.
- Brute force over all 210 combinations confirms the optimum.

Stage 3 — 6-qubit outcome simulation

One qubit per customer, RY angle from its probability

- Qubits: C0031; C0079; C0256; C0338; C0170; C0327 — qubit i carries customer i,

RY(2·arcsin(√p)) so P(|1⟩) = p. Customers assumed independent.

- P(≥3 subscribe): quantum 0.3574 (16,384 shots, Aer)

exact Poisson-binomial 0.3603 • Monte Carlo (200k) 0.3595

- Quantum − exact error: -0.0028 • noisy-Aer estimate 0.3787
- Transpiled circuit depth 2 — 6 RY gates + 6 measurements, no entangling gates,

Error analysis — what can go wrong, and how big it is

Honest limits, measured not asserted

- Stage 1 is the bottleneck: only 35 positive examples in 300 train rows, so ranking (PR-AUC ~0.20–0.23) is modest — and every later stage inherits those probabilities unchanged.
- Stage 2: QAOA is heuristic; the brute-force check bounds its gap. An infeasible selection would earn zero — ours is feasible.
- Stage 3: finite-shot error (~±0.004 at 16,384 shots) plus readout / depolarising noise in the noisy run; both reported next to exact.

- No quantum-advantage claim: Aer simulator, small problem, classical references beat or match the quantum steps — the quantum parts are demonstrated, validated, and kept separate from classical baselines.

Business impact — what the bank should do

Contact these 6, expect about half the target to convert

- Contact C0031, C0079, C0256, C0338, C0170, C0327 — total cost 10 of a 25 budget, capacity 15 of 22. Expected net value 33.22.
- Chance at least 3 subscribe: ~36% — the campaign target is a coin-flip, not a certainty; do not over-commit follow-up.
- Next: re-score weekly as new outcomes arrive, refit on train only, and re-run the same pipeline — the selection step is the cheap part.
- Do not contact the other top-10 names: they cost more per expected deposit than the six chosen once budget and capacity bind.

Reproducibility — run it yourself

Everything is seeded, pinned, and runnable

- .venv/bin/python run_pipeline.py — regenerates every result file (stage1_predictions.csv, stage2_selection.csv/.json, stage3_simulation.csv/.json) from resources/ in one command.
- Seeds: 42 for data split, model CV, QAOA sampling, Aer shots, transpiler, and Monte Carlo. Versions pinned in requirements.txt.
- Backends: AerSimulator only (no IBM Quantum account, no hardware). QAOA via Qiskit Optimization + Qiskit Algorithms; kernel via Qiskit Machine Learning FidelityQuantumKernel.

- Repo: public GitHub, MIT licence, README is the technical report.